

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

May 2014

EC210 Analog Communication

Date: 09.05.2014, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Attempt any three.

[06]

1. Define modulation index of an amplitude modulation (AM) signal
2. Draw the frequency spectrum of amplitude modulation (AM) wave.
3. Explain basic principle of diode detector in amplitude demodulation.
4. Derive the mathematical equation of modulation index in terms of P_{total} and P_c .
5. An audio frequency signal $10 \sin 2\pi 500t$ is used to amplitude modulate a carrier of $50 \sin 2\pi \times 10^5$. Calculate i) modulation index ii) amplitude of each sideband frequencies iii) total power delivered to the load of 600Ω .

Q - 2 (a) List the method of suppression of unwanted sideband and compare them.

[08]

(b) Draw and explain FET reactance modulator used for frequency modulation.

[07]

OR

Q - 2 (a) Write a note on

[08]

i) Independent sideband system

ii) Vestigial sideband transmission

(b) Explain Armstrong frequency modulation system with the help of block diagram.

[07]

Q - 3 (a) Compare phase modulation with amplitude modulation.

[07]

(b) With the help of neat circuit diagram and phase diagram explain the working of Foster seely discriminator.

[07]

OR

Q - 3 (a) How directly modulated frequency modulator transmitter utilized in automatic frequency control (AFC)?

[07]

(b) Explain basic method of obtaining frequency modulation from phase modulation.

[07]

SECTION – II

Q - 4 State all the fourier transform properties and Prove any three. [07]

Q - 5 (a) What do you mean by Energy Spectral Density? Prove that fourier transform of autocorrelation of energy signal is Energy spectral density. [07]

(b) Define the term Noise factor and Noise figure. Derive friss formula for noise factor. [07]

OR

Q - 5 (a) What do you mean by Power Spectral Density? Prove that fourier transform of autocorrelation of power signal is Power spectral density. [07]

(b) Derive friss formula for Noise temperature. Cable has an insersion loss of 2 dB. Determine its equivalent noise temperature referred to the input. [07]

Q - 6 (a) What do you mean by Noise? List out all types of noise and explain thermal noise. [07]

(b) Explain Super heterodyne receiver in detail [07]

OR

Q - 6 (a) Define Tuning and Tracking and Explain Automatic Gain Control in detail. [07]

(b) How noise affects receiver? A receiver has a noise figure of 12 dB and it is fed by LNA that has a gain of 50 dB and a noise temperature of 90K. Calculate the noise temperature of the receiver and the overall noise temperature of the receiving system. [07]

OR

OR